EXPLICIT HAMILTONIAN BALL PACKINGS: OVERVIEW AND READER'S GUIDE

LIANG SONG

Abstract. One explicit smooth compactly supported Hamiltonian packs the prescribed closed unit balls arbitrarily densely into the four-ball for every $k \geq 10$ and every positive volume deficit. A linear rearrangement identity and a $\mathbb{Z}/3$-equivariant return from the Fermat cubic make the construction uniform, including all non-square k. The series also gives a new constructive proof of Biran's $k \geq 9$ theorem, general-capacity and higher-dimensional packings, a halting algorithm with finite polynomial-cutoff output, elementary density barriers, and ODE-free C^4 packing maps. This guide locates the main statements, proof ideas and reproduction methods, distinguishing the Lean-verified Hamiltonian theorems from the direct-map proofs and numerical implementations.

1. Results and reading route

Uniformly for every integer $k \geq 10$—including all non-square values $k = 10, 11, 12, \ldots$—and every volume deficit $\varepsilon \in (0, 1)$, this project constructs an explicit smooth compactly supported Hamiltonian on $\mathbb{R} \times \mathbb{R}^4$ whose time-one flow carries k disjoint closed unit balls into the four-ball of occupied volume fraction $1 - \varepsilon/4$, extends the construction to $k = 9$, unequal capacities, higher dimensions, and a halting finite-output algorithm, analyses the sharp limits of elementary toric/window packings, and constructs ODE-free closed-form C^4 packing embeddings with exact-real inverse certificates. Each of the four papers has a distinct scope:

	Reading route	Proof / reproduction
I	Equal balls in B^4: start with Theorem A, then the linear comparison identity in §4 and Hamiltonian assembly. Robustness and RK4 containment are Proposition B and Corollary D.	`paper/p1`, supplement; Lean 4
II	General capacities, Biran's $k \geq 9$ theorem, and higher dimensions: Theorem 1; finite-output algorithm: Theorems 5.2–5.3.	`paper/p2`, supplement; Lean 4
III	Elementary toric/window constructions and their sharp, class-specific density barriers.	`paper/p3`; `solution_epoch.py`
IV	ODE-free C^4 maps and inverse certificates: Theorem 1.1, §§2–7. Radial-collar generator: §8.	`paper/p4`; `solution_embedding.py`

Biran [Bir97] proved existence of arbitrarily dense equal-ball packings of B^4 for every $k \geq 9$; classical explicit full packings include the square-number constructions of Traynor and Schlenk [Tra95, Sch05], alongside folding, wrapping, tunnelling, singular-curve/polarisation, and almost-toric constructions. Papers I, II, and IV build on the twisted disc-bundle framework introduced in dimensions $2n \geq 6$ by OpenAI [OAI26], introducing new four-dimensional ingredients that avoid J-holomorphic curves, Seiberg–Witten theory, and contraction fixed points.

Version 1.2.0; reader's guide to Papers I–IV in `https://github.com/s-JoL/Symplectic-Ball-Packing` (all versions: doi:10.5281/zenodo.23756575).

2. Principal statements and regularity

Write $B(c) = \{\pi|z|^2 \le c\} \subset \mathbb{R}^{2n}$ and $B^\circ(c) = \{\pi|z|^2 < c\}$, so $B(s,\pi)$ is the closed Euclidean unit ball centred at s. Regularity is kept distinct across the papers: Papers I–II construct globally C^∞ compactly supported Hamiltonians on $\mathbb{R}\times\mathbb{R}^{2n}$, whereas Paper IV constructs ODE-free C^4 symplectic embeddings on an open ball $\mathbb{B}^\circ(r_+)$ ($r_+ > 1$).

Paper I ($k \ge 10$ equal balls and RK4 containment).

(i) *Theorem A (`four_dim_theorem`).* For every integer $k \ge 10$ and $\varepsilon \in (0,1)$, with $R_Q = (k/(1-\varepsilon/4))^{1/4}$ and $s_i = (3i-3,0,0,0)$ ($1 \le i \le k$), the explicit Hamiltonian $H_{k,\varepsilon} \in C_c^\infty(\mathbb{R}\times\mathbb{R}^4)$ has a global flow whose time-one map carries the k closed unit balls $B(s_i,\pi)$ onto pairwise disjoint subsets of $B^\circ(\pi R_Q^2)$, occupying the volume fraction $k/R_Q^4 = 1-\varepsilon/4$. Two four-dimensional ideas drive the short proof: for $K(p) = (3\sqrt{k}-9)p$ linear, the profile defect $K-P$ on $[0,t_0/3]$ is an affine multiple of the generally asymmetric tent on $[0,h]$, $h = t_0/3$, with vertex $a = t_0/\sqrt{k}$ and equal endpoint values h. Its reciprocal slopes sum to one, so its increasing rearrangement is the identity; and $\mathbb{Z}/3$-equivariance $Z_0 \mapsto e^{2\pi i/3}Z_0$ of the Fermat-cubic return keeps the balls off the line at infinity $\{Z_0 = 0\}$ without C^0 Moser estimates.

(ii) *Proposition B and Corollary D (`four_dim_stability`, `epoch_rk4_convergence`).* Put $R = (k/(1-\varepsilon/2))^{1/4}$ and $g = R - R_Q > 0$. For any space-time Lipschitz and supremum bounds $K, C \ge 0$ of $X_{H_{k,\varepsilon}}$ on $[0,1]\times\mathbb{R}^4$, set $M = K + KC$, $\delta_0 = g/(4e^K) > 0$, and $N_0 = \lceil 4Me^K/g\rceil + 1$. Then for every $N \ge N_0$, every $x \in B(s_i,\pi)$, and every N-step sequence $(y_m)_{m=0}^N$ with $|y_0 - x| \le \delta_0$ and per-step state perturbation $|y_{m+1} - \mathrm{RK4}(H_{k,\varepsilon}, m/N, 1/N, y_m)| \le \delta_0/N$, one has $|y_N| \le R$ ($k/R^4 = 1-\varepsilon/2 > 1-\varepsilon$). Here K and C come from smoothness and compact support and are not numerically instantiated, so Corollary D gives no executable step or precision budget and does not certify a `float64` program or nearby-pair symplecticity test.

Paper II (General capacities, higher dimensions, and finite-output algorithm).

(i) *Theorem 1 and Corollaries 2–5 (`main_theorem`, `neighbourhood_criterion`, `equal_balls`, `epoch_theorem`).* One named $C_c^\infty(\mathbb{R}\times\mathbb{R}^{2n})$ Hamiltonian realises every packing with $\sum_i c_i^2 < C^2$ and $3c_i < C$ in $\mathbb{R}^4$ (via a closed-form scalar mollifier comparator, giving Biran's full-packing theorem for all $k \ge 9$ equal balls without J-holomorphic curves or Seiberg–Witten theory) and every packing with $\sum_i c_i^n < C^n$ and $c_i + c_j < C$ in $\mathbb{R}^{2n}$, $n \ge 3$ (making every step of [OAI26] explicit, with a finite rational contraction iterate and separate small-ball and one-large-ball returns).

(ii) *Theorems 5.2–5.3 (`original_name_algorithm`, `epoch_name_algorithm`).* One uniform halting oracle machine, given rational approximations to admissible data, outputs a rational polynomial P and cutoff radius B whose smooth compactly supported Hamiltonian $\beta_B \cdot P$ satisfies the packing specification (for the Epoch problem, $k \ge 9$ and a name for $\varepsilon \in (0,1)$, packing at density $1-\varepsilon/2$).

Paper III (Elementary constructions and density barriers). Constructs explicit toric and window embeddings packing $n(n+1)$ open balls of capacity $\lambda < 1$ into the open ball $B^\circ(n+\frac{1}{2})$ (written $B(A)$ in Paper III), every k at density at least $1 - 1/\sqrt{k}$, and ten balls at density $360/361$ ($A = \frac{19}{6}$). It then proves three sharp obstructions in its elementary configuration class: unimodular lattice triangles leave a positive density gap for every non-square k; 1-periodically balanced profiles satisfy the Newton barrier $A^2 \ge k + \alpha^2$, where $\alpha = \mathrm{dist}(A,\mathbb{Z})$; and for $k = 10$, any configuration with eight or more full-height columns satisfies $A \ge \frac{19}{6}$.

Paper IV (ODE-free closed-form C^4 embeddings and inverse certificates).

(i) *Theorem 1.1 and Lemma 7.4.* For every $k \geq 10$ and $\varepsilon \in (0,1)$, explicit 11-stage compositions $f_0, \ldots, f_{k-1} \colon \mathbb{B}^\circ(r_+) \to \mathbb{B}^\circ(R_{\mathrm{req}})$ ($r_+ > 1$, $R_{\mathrm{req}} = (k/(1-\varepsilon))^{1/4}$) built from elementary charts, generating-function keyholes (solved by scalar 1D monotone roots), C^4 deck-equivariant toric twists, and a cancellation-free moment-simplex transfer are C^4, injective, symplectic, and pairwise disjoint, and Preimage(j, Y) is an exact left inverse certifying non-membership whenever $Y \notin f_j(\mathbb{B}(1))$. All parameter hypotheses, grouped as (H0)–(H15) and the band-separation condition, hold analytically for all $k \geq 10$ and $\varepsilon \in (0,1)$.

(ii) *Proposition 8.1.* Gives a normalised Liouville primitive S_i on $\mathbb{B}^\circ(r_+)$ and the radial–Alexander generator $G_t^{(i)}(y) = -(\dot{v}/v^3)(2(S_i(w) - S_i(a_i)) + \omega_0(a_i, w))$, and exact/quantitative ODE trajectory certificates ($\phi_H^t(s_i + u) = \gamma_t^{(i)}(u)$ on $[0,1]$ iff $\partial_t \gamma_t^{(i)}(u) = X_H(t, \gamma_t^{(i)}(u))$ on $[0,1]$, with Grönwall endpoint bound δe^L; in Lean, `picard_lindelof_curve_unique` uses Lipschitzness alone while `picard_lindelof_epoch_certificate` assumes $H \in C^\infty$). The formula describes the radial collar stage; lifting these direct maps to a standalone global Hamiltonian flow (distinct from Paper I's C_c^∞ flow) is left to future work, and the abstract Lean ODE lemmas do not certify such a lift.

3. Logical dependencies across the series

- **Papers I and II (coupled formalised core):** Paper I is self-contained for $k \geq 10$ equal balls in dimension four; Paper II quotes the shared four-dimensional handle and cubic-return lemmas from Paper I (`p2/p1refs.tex`) and supplies the general comparator, higher-dimensional models, and finite-output algorithm. In Lean 4, one library `SymplecticPacking/` serves both papers: Paper I's $H_{k,\varepsilon}$ is the specialisation of Paper II's `General.packingHamiltonian` via `four_dim_data`, and `verification/RK4Convergence.lean` imports `Statement.lean` without modifying any of the 525 Lean source files in `SymplecticPacking/`.
- **Paper III (independent elementary analysis):** Mathematically independent of Papers I, II, and IV; explains which (k, ε) regimes admit gentle toric/window flows and motivates `program/solution_epoch.py`.
- **Paper IV (independent closed-form direct maps):** Mathematically self-contained and proved directly in real-analytic/generating-function charts without invoking Theorem A of Paper I or any ODE flow; provides the exact-real proof for `program/solution_embedding.py`.
- **Independence from J-holomorphic curves:** Every constructive theorem in Papers I, II, III, and IV is proved from explicit maps, differential forms, and elementary analysis alone; only the necessity direction of Paper II, Corollary 2 invokes Gromov's classical two-ball obstruction. The audit in `verification/Audit/Independence.lean` checks a stated exclusion list of named dependencies across the full Lean proof-term closure.

4. Verification layers and accompanying programs

1. **Lean 4 kernel and fresh replay (Papers I–II and `RK4Convergence.lean`):** In `verification/Problem.lean`, phase space, ω_0, X_H, admissibility, and `PackingSpec` are defined from Mathlib alone, quantifying over *every* integral curve from *every* point of *every* source ball. All main theorems in `Statement.lean`, `Computability.lean`, `Epoch Computability.lean`, and `RK4Convergence.lean` pass `#assert_publication_axioms` (`propext`, `Classical.choice`, and `Quot.sound` only; no `sorryAx` or `native_decide`) and are re-checked from scratch in a fresh kernel process (`scripts/check_kernel.sh`).
2. **Syntactic explicitness and uniformity metaprograms:** `#check_explicit` audits the non-proof definition closure of `packingHamiltonian` and `epochHamiltonianQ`; `#check_constructive` checks that the halting oracle machines contain no classical branching.

3. **Human-readable proofs and cross-review (Papers I–IV):** Statements in Papers I–II carry source tags indexed in `verification/PAPER-INDEX.md`; Papers III–IV are standalone mathematical proofs reviewed across multiple independent models.
4. **Programs and Epoch problem mapping (`program/` and `EPOCH.md`):** Two official Python generators accompany the papers as numerical corroboration (neither is formalised in Lean): `solution_epoch.py` passes documented sampled fixed-step `float64` RK4 and per-call timing checks in an elementary (k, ε) regime (with finite smoothness at the keyhole origin and time junctions, exterior branch rays, and for $k = 10$ a piecewise C^2 window seam), and `solution_embedding.py` evaluates Paper IV's closed-form C^4 maps and exact-real inverses for all $k \geq 10$, $\varepsilon \in (0, 1)$, checked in `float64` (`check_embedding.py`) and AST-transformed 100-digit `mpmath` (`check_embedding_mp.py`). An unfinished prototype exploring a multi-stage global Hamiltonian lift of Paper IV's chain is archived separately in `experiments/hamiltonian_lift/` with known collar sheet-jump reproducers and is not part of the certified deliverables. See `REVIEW.md` for the formal audit guide and `EPOCH.md` for an item-by-item comparison with Epoch AI's problem specification [Epo26].

Authorship and the role of AI

The manuscripts, mathematical proofs, Lean 4 formalisation and accompanying programs of this release were generated by AI models, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic), Gemini 4 Carbon (Google) and other models. Liang Song chose the problem, designed and directed the research environment, organised cross-model review and takes sole responsibility for the release.

Trustworthiness rests on explicit verification layers: the main theorems of Papers I–II (including the finite-output algorithm and perturbed fixed-step RK4 target containment) are checked in Lean 4 against an independent specification importing only Mathlib, using only the three standard axioms and re-verified by fresh kernel replay; Papers III–IV give human-readable proofs supported by multi-model cross-review and numerical checks; and program outputs serve only as numerical corroboration.

References

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[Tra95] L. Traynor, *Symplectic packing constructions*, J. Differential Geom. **42** (1995), no. 2, 411–429.

Email address: `sl12160010@gmail.com`